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Q5. Procedure registermember (name[]: string, ID: int, membertype: int){
Output
Output "enter your name ad student ID"
Input name[], ID
Outpt "Are you student, or faculty?"
Input membertype
if membertype != NOT 1 AND membertype NOT 0
}

re-prompt using while loop

Procedure displaymenu ( )
Output "1. Member registation, 2. Display available books, 3. Borrow a book, 4. Return a book, 5. Display member record, 6. Exit"
}

Procedure displaybooks (bookid[]: array of int, booktitles[]: array of string, availc[]: array of int, totalbooks: int)
{
for (i = 1 to totalbooks
Outpt bookid[i], booktitles[i], availc[i]
}

Procedure borrowbook(bookid[]: array of int booktitles[]: array of string, availc[]: array of int, totalbooks: int, borrowedbooksid[]: array of int, borrow, membertype: integer)
{
l/d: int, id←0, index, days: int ← 0

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valid ← 1, boolean ← false
Output "bookid of book u wanna borrow?"
Input id
for i ← 0 to totalbooks
    if bookid[i] = id
        valid ← true
        index ← i

if NOT(valid)
    Output "Incorrect id"

if borrow >= 3
    Output "you can't borrow any more books"

if available[index] <= 0
    Output "This book isn't available"

how
Output "how many days do you want to keep the book"
Input days. if membertype = 0
if days <= 0 OR days > 14.
    Output "invalid days"
else if membertype = 1
    if days <= 0 OR days > 30
        Output "invalid days"

borrowedbookid[borrowed c] = id;

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borrow ← borrow + 1
available[index] ← 1 }

function returnbook (bookid[]: array of integer, in
avail[]: array of int, totalbooks: int, borrowbookid[]:
array of int, borrow, membertype: int, totalfree: real)
{Return Real {
id: int ← 0 borrow_index
days, index, extra: int ← 0
valid: boolean ← false
code[8]: array of char
Output "Enter book id:"
Input id
while days <= 0
Output "how many days did you keep it?"
Input days

for i = 1 to borrow {
if borrowbookid[i] == id
valid ← true
borrow_index ← i

if NOT (valid)
Output "incorrect id"

for j = 1 to totalbooks
if bookid[i] == id
index = i

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```
overallc[index] += 1
currentfine ; real ← 0

if  membertype = 0
    if days > 14
     extra = days - 14
     currentfine = extra * 20
     totalfine += currentfine
     Output "Enter membership code:"
     Input code
     If code = "LIB2026"
        currentfine = currentfine * 0.30

else if  membertype = 1
    if days > 30
     extra = days - 30
     currentfine = extra * 10
     totalfine += currentfine
     Output "Enter membership code:"
     Input code
     if code = "LIB 2026"
        currentfine = currentfine * 0.30

for k = borrowindex to (borrowc - 1)
    borrowedbookid[i] = borrowedbookid[i+1]
borrowedbookid[borrowc] = 0
borrowc -= 1
}
```

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```
Procedure displaymemberrecord ( name : string, studentId,
memberType, borrowedc : int, borrowbookId : array of int,
totalfine, fine afteraaiver : Real) {
Output name, idstudentid, membertype
if borrowc > 0
    Output :
    for i = 1 to borrowc
    Output borrowbookId[i]

Output borrowedc, totalfine, fineafteraiver }
```

→ main implementation

```
choice, studentid, membertype, borrowc, totalbooks, total
fine, payblefine : int ← 0
bookId[5] = {101, 102, 103, 104, 105} (array of int)
booktitles[5] : array of string ← { "C Programming" "Data Structs", "Calc", "Physics",
"Discrete math" }
copies avail[5] : array of int ← {3, 2, 1, 4, 2}
borrowedbookid[3] = {0, 0, 0} (array of int)
name : string
While choice NOT(6)
    displaymenu()
    Output "Enter choice:"
    Input chola
    case of choices
      1 : registermember(name, studentid, membertype)
      2 : displaybooks (bookId, booktitles, avail, totalbooks)
      3. borrowbook (bookId, booktitles, availc, totalbooks
```

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borrowedbookId , borrow , memberType

4. payableFine += returnBook(bookId, availableBooks, totalBooks, borrowBookId, borrow, memberType, totalFine)

5. display memberRecord(name, studentId, memberType, borrowedBooksId, borrow, totalFine, payableFine)

~~Other~~
Otherwise
Output Invalid choice

→ Inputs :
1. Choice
2. name
3. StudentID
4. Membertype (0 or 1)
5. ~~borrow~~ bookId
6. days (days kept too)
7. membership code

→ Output:
1. menu choices
2. name
3. ID
4. membertype
5. bookIds of all borrowed books.
6. Total books, 7. total fine, 8. fine payable.

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